SpatialEntail Problem Generation

Idea: choose what the question should establish, work backward to build its premises, then check what those premises actually imply.

Inputs: question type, desired number of possible answers, reasoning length, and number of distractors.

Algorithm 1: Generate one verified spatial reasoning problem

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for each attempt within the retry budget do
    Choose object names
    Build a target conclusion or a starting set of possible answers
    path ← sample a valid rule sequence for the requested length
    goal ← a statement needed to establish the target
    for rule in reverse(path) do
        Replace goal with the statement needed before applying this rule
        Add any supporting premise required by the rule
    end
    Add the final goal as a premise; add distractors and shuffle
    Solve the resulting question and check its proof
    if the question does not meet the requested constraints then
        Reject this candidate and try again
    end
    Render and validate the question, options, answer, and solution
    return the verified example
end
return generation failure
```

Valid rule sequences are sampled uniformly before candidate rejection; the accepted examples need not be uniform. The solver determines the final answers, rather than trusting the generator's intended result.